

# Regional Satisfaction Survey Codebook

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# Regional Satisfaction Survey Codebook

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## Project Overview: Summary of Project Details

**Project Title:** PDF Codebook with Sample Data

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**Required Citations:**

Nathanael Rosenheim. (2025). npr99/pypdfcodebook: v0.5.1 Zenodo Release (v0.5.1). Zenodo. <https://doi.org/10.5281/zenodo.17956283>

**Project Description :** pyPDFCodeBook helps researchers and data professionals create clear, attractive codebooks for tabular datasets. Codebooks document essential metadata—project descriptions, data provenance, variable definitions, and summaries—ensuring your data is understandable and reproducible. While datasets contain values, they rarely explain what each column or row represents. Codebooks fill this gap by providing structured, self-explanatory documentation. They reinforce best practices such as tidy data, unique keys, and transparent variable origins. With pyPDFCodeBook, generating a professional, easy-to-read codebook takes just a few clicks—laying the foundation for good data science and reproducible research. Remember: You are your number one data user, so help your future self out and document your metadata.

**Data Provenance :** All Regional Satisfaction Survey sample data generated using Claude Sonnet 4 in VS Code Agent mode. All values are random and do not represent actual survey data. Logo image generated using MS Copilot.

**Funding Source: Award name and number:**

Center for Risk-Based Community Resilience Planning : NIST-70NANB20H008, NIST-70NANB15H044 (<http://resilience.colostate.edu/>)

Southeast Texas Urban Integrated Field Lab: DE-SC0023216 (<https://setx-uifl.org/>)

**Related Works:**

**Keywords:** Codebook, data dictionary, metadata

**License:** Open Data Commons Attribution License (<https://opendatacommons.org/licenses/by/summary/>)

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## Data Dictionary: Summary of Variables

| Variable Name | Data Type | Length | Categorical | Variable Label            |
|---------------|-----------|--------|-------------|---------------------------|
| rid           | String    |        |             | Random Survey Response ID |
| region        | Int       |        | True        | Geographic Region         |
| satscore      | Int       |        | True        | Satisfaction Rating       |
| weight        | Float     |        |             | Survey Weight             |
| age           | Int       |        |             | Age in Years              |

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## Variable Details and Notes

The following pages provide details on each variable. Where applicable, notes provide links to verify data. Categorical variables include details on category codes.

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rid: Random Survey Response ID

| Variable characteristic | Variable details |
|-------------------------|------------------|
| variable type           | string           |
| total cases             | 1,000            |
| valid cases             | 1,000            |
| missing cases           | 0                |
| unit of measure         | Responses        |
| unit of analysis        | Survey response  |
| unique values           | 1,000            |
| minimum length          | 5                |
| maximum length          | 5                |
| example 1               | M1101            |
| example 2               | E1526            |
| example 3               | P1826            |
| example 4               | S1226            |

Variable Notes: rid

1. Unique, non-missing key for sample data.
2. Randomly generated 5 digit alphanumeric string identifier.
3. Range from A1000 to Z9999.

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region: Geographic Region

| Variable characteristic | Variable details                     |
|-------------------------|--------------------------------------|
| variable type           | categorical (Int)                    |
| categorical type        | nominal                              |
| total cases             | 1,000                                |
| valid cases             | 1,000.0                              |
| missing cases           | 0                                    |
| weighted total cases    | 4,700.00                             |
| unit of measure         | Responses by Region                  |
| unit of analysis        | Survey response                      |
| range                   | minimum value: 1 to maximum value: 4 |

region: Geographic Region - Categorical codes, labels and frequencies

| Code | Label    | Count of Responses | Percent Responses | Weighted Sum | Weighted Percent |
|------|----------|--------------------|-------------------|--------------|------------------|
| 1    | 1. North | 200                | 20.00%            | 2,000.0      | 42.55%           |
| 2    | 2. South | 300                | 30.00%            | 1,200.0      | 25.53%           |
| 3    | 3. East  | 400                | 40.00%            | 300.0        | 6.38%            |
| 4    | 4. West  | 100                | 10.00%            | 1,200.0      | 25.53%           |

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satscore: Satisfaction Rating

| Variable characteristic | Variable details                     |
|-------------------------|--------------------------------------|
| variable type           | categorical (Int)                    |
| categorical type        | ordinal                              |
| total cases             | 1,000                                |
| valid cases             | 900.0                                |
| missing cases           | 100                                  |
| weighted total cases    | 4,222.25                             |
| unit of measure         | Satisfaction Level                   |
| unit of analysis        | Survey response                      |
| range                   | minimum value: 1 to maximum value: 5 |
| mean                    | 3.67                                 |
| weighted mean           | 3.75                                 |

satscore: Satisfaction Rating - Categorical codes, labels and frequencies

| Code | Label                | Count of Responses | Percent Responses | Weighted Sum | Weighted Percent |
|------|----------------------|--------------------|-------------------|--------------|------------------|
| 1    | 1. Very Dissatisfied | 38                 | 4.22%             | 173.75       | 4.12%            |
| 2    | 2. Dissatisfied      | 110                | 12.22%            | 443.0        | 10.49%           |
| 3    | 3. Neutral           | 174                | 19.33%            | 755.25       | 17.89%           |
| 4    | 4. Satisfied         | 364                | 40.44%            | 1,725.75     | 40.87%           |
| 5    | 5. Very Satisfied    | 214                | 23.78%            | 1,124.5      | 26.63%           |

Variable Notes: satscore

1. Five-point Likert scale measuring overall satisfaction.
2. Missing values indicate participant did not respond to this question.



# Regional Satisfaction Survey Codebook

weight: Survey Weight

| Variable characteristic | Variable details                            |
|-------------------------|---------------------------------------------|
| variable type           | numeric (Float)                             |
| total cases             | 1,000                                       |
| valid cases             | 1,000                                       |
| missing cases           | 0                                           |
| unit of measure         | Population                                  |
| unit of analysis        | Survey response                             |
| range                   | minimum value: 0.75 to maximum value: 12.00 |
| mean                    | 4.70                                        |
| median                  | 4.00                                        |
| standard deviation      | 4.17                                        |
| 10th percentile         | 0.75                                        |
| 25th percentile         | 0.75                                        |
| 50th percentile         | 4.00                                        |
| 75th percentile         | 10.00                                       |
| 90th percentile         | 10.20                                       |

## Variable Notes: weight

1. Statistical weight for population inference and representative results.
2. Used to adjust for sampling bias and ensure results reflect target population.
3. Based on sample design for each region.
4. Region 1 weight = 10.0, Region 2 weight = 4.0, Region 3 weight = 0.75, Region 4 weight = 12.0.
5. Regions 1, 2, and 4 are under sampled regions. Region 3 is over sampled.

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age: Age in Years

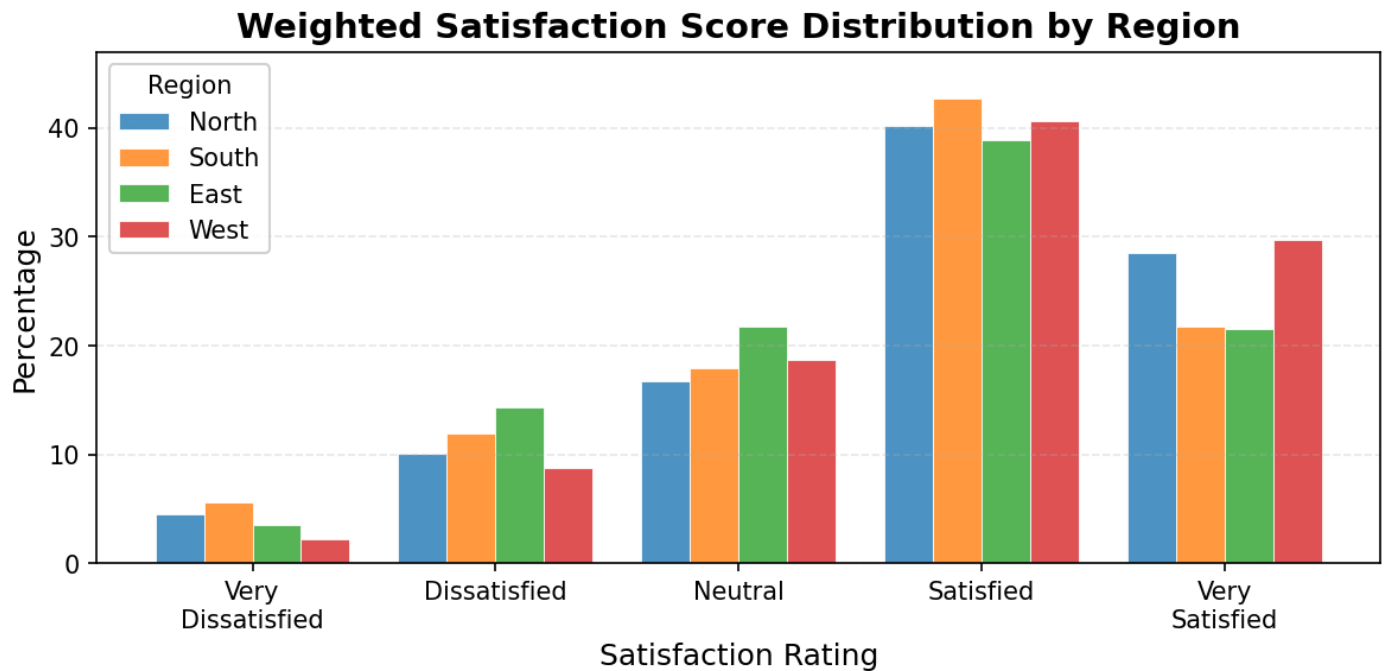
| Variable characteristic | Variable details                             |
|-------------------------|----------------------------------------------|
| variable type           | numeric (Int)                                |
| total cases             | 1,000                                        |
| valid cases             | 1,000                                        |
| missing cases           | 0                                            |
| unit of measure         | Years                                        |
| unit of analysis        | Survey response                              |
| range                   | minimum value: 18.00 to maximum value: 99.00 |
| mean                    | 47.03                                        |
| median                  | 37.00                                        |
| standard deviation      | 21.86                                        |
| 10th percentile         | 23.00                                        |
| 25th percentile         | 32.00                                        |
| 50th percentile         | 37.00                                        |
| 75th percentile         | 59.00                                        |
| 90th percentile         | 81.00                                        |

## Variable Notes: age

1. Age of survey respondent at time of survey.
2. Range from 18 to 99 years.
3. Missing values indicate participant did not provide age information.
4. Surveys completed between November 2025 and December 2025.

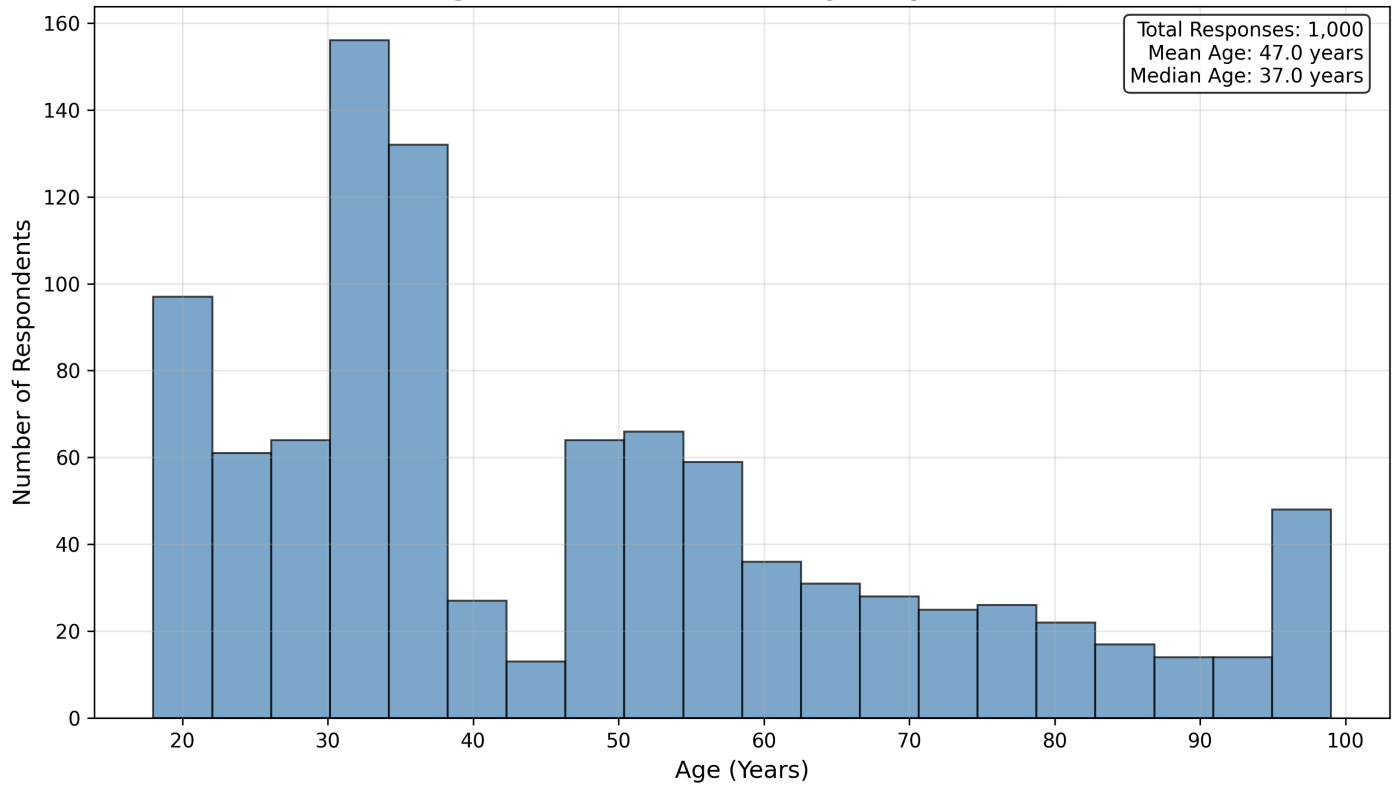
# Regional Satisfaction Survey Codebook

## Figures



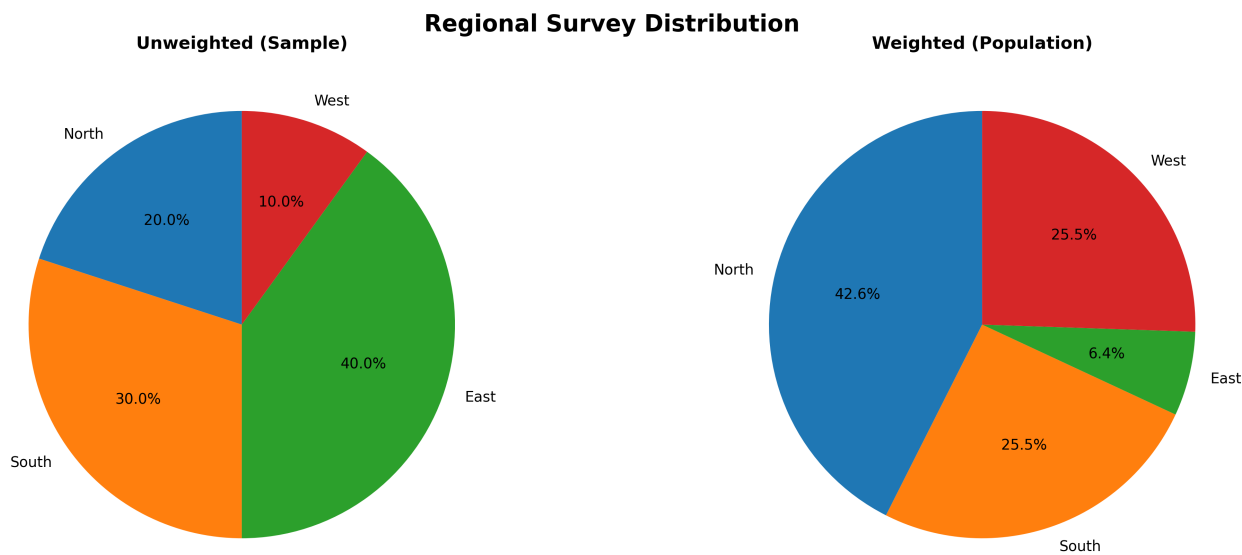
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## Age Distribution of Survey Respondents



Provenance: pdfcb\_00c\_sampledata.csv | 2025-12-17 14:54:51 | Made with the help of Claude Sonnet4 in VS Code Agent Mode

# Regional Satisfaction Survey Codebook



Provenance: pdfcb\_00c\_sampledata.csv | 2025-12-17 14:46:06 | Made with the help of Claude Sonnet4 in VS Code Agent Mode

# Regional Satisfaction Survey Codebook

## Key Terms and Definitions

**Codebook:** "A codebook describes the contents, structure, and layout of a data collection... The main body of a codebook contains unambiguous variable level details. " (ICPSR: <https://www.icpsr.umich.edu/sites/icpsr/posts/shared/what-is-a-codebook>).

**Data Dictionaries:** A compact list of the variables in the dataset. Includes variable name, data type, length, if the variable is categorical or not, and the variable label.

**Key:** "A key is a variable or set of variables that uniquely identifies the elements of a table. The variables that form the key never take on missing values, and a key's value is never duplicated across rows of the table." (Gentzko & Shapiro (2014). Code and Data for the Social Sciences: A Practitioner's Guide. p. 20 <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>)

**Metadata:** "Metadata is information about the data, descriptors that facilitate cataloguing data and data discovery." (The Turing Way. "Metadata in Research Data Management." The Turing Way Book: <https://book.the-turing-way.org/reproducible-research/rdm/rdm-metadata>).

**Provenance:** "The precise subset and version of data a set of result originates from. (The Turing Way Book: <https://book.the-turing-way.org/reproducible-research/vcs/vcs-data/>).

**Tabular Dataset:** "Datasets made up of rows and columns." (Tidy data: <https://tidyr.tidyverse.org/articles/tidy-data.html>)

**Tidy Data:** "Each variable is a column; each column is a variable. Each observation is a row; each row is an observation. Each value is a cell; each cell is a single value." (Tidy data: <https://tidyr.tidyverse.org/articles/tidy-data.html>)

**Unit of Observation:** "The object(s) of analysis for the data collection, such as an organization, individual, or household." (ICPSR: [https://icpsr.github.io/metadata/icpsr\\_study\\_schema/#unit\\_of\\_observation](https://icpsr.github.io/metadata/icpsr_study_schema/#unit_of_observation)).